F481 — MOVER 1 theorem layer: the conformal Casimir IS the FORCED mass operator on the SO(4,2) arena (a candidate THIRD structural bank — operator pinned, not fitted), the mass is LINEAR in it (F292, not the AdS/CFT m^2 — flagged honestly), and $\Delta = \hat{D} + d$ is the ONE open forward gate (the SO(4,2) discrete-series descent). THE FORCING CHAIN (operator): (1) the arena SO(4,2) is FORCED by the banked descent (SO(5,2)→SO(4,2)→SO(3,1), emergent 4D the read-out arena); (2) $\Delta(\Delta-d)$ is THE quadratic Casimir of SO(4,2) — the unique degree-2 invariant that labels a rep — and mass is a rep-invariant scalar (it labels the particle's rep), so mass = the rep Casimir = $\Delta(\Delta-4)$ [forced up to the reading]; (3) the reading is mass ∝ Casimir LINEAR (F292's established linear-energy result), NOT m^2 ∝ Casimir (standard AdS/CFT). So 'masses = the linear SO(4,2) conformal Casimir $\Delta(\Delta-4)$' is FORCED structure — the operator is pinned by (arena forced) + (Casimir = the unique rep-invariant) + (F292 linear), not chosen to fit. This is a candidate THIRD STRUCTURAL BANK (arena + operator), alongside Peirce (T2511) and the descent closure — gated only on the Δ VALUES, not the operator. HONEST m-vs-m^2 FLAG (checked against data): Grace's map $\hat{D}$={5,2,0} (F446 KW strata, target-innocent) → $\Delta = \hat{D}+d$ = {9,6,4} → Casimir {45,12,0}. The DATA picks LINEAR: m_b/m_s = 45 directly (not $\sqrt{(45/12)}$=1.94), so BST uses mass ∝ Casimir (F292 linear), a BST result DISTINCT from standard AdS/CFT's m^2 ∝ Casimir — flagged, not hidden. THE ONE OPEN GATE: is $\Delta = \hat{D} + d$ FORCED? $\hat{D}$={5,2,0} is the pre-existing target-innocent KW-strata spectrum (F446); the map $\Delta=\hat{D}+d$ has ZERO free parameters (d=4 forced) and hits 2 targets + 1 massless bonus (not form-cheap). But WHY $\Delta=\hat{D}+d$ (affine, +d)? Must come forward from the SO(4,2) discrete-series descent of the D_IV^5 K-types (the conformal dimension of the descended operator = its stratum data + the boundary d) — the discrete-series/defect-CFT relation to DERIVE (Grace's lead), NOT fabricated from memory. WHEN $\Delta=\hat{D}+d$ is forward: the different-factors question DISSOLVES (both factors = one Casimir at forced strata), the down-ladder banks 8→10 (with the deposit 5/3 combination checked), and the operator-pinning banks as structure. VERDICT: operator FORCED (candidate 3rd structural bank); $\Delta=\hat{D}+d$ = the one open forward gate; deposit + m-vs-m^2 flagged honest. No observable bank yet, count 8. For Grace (MOVER 1: derive $\Delta=\hat{D}+d$ from the discrete-series descent — the whole gate), Keeper (candidate 3rd structural bank: operator forced; adjudicate), Elie (deposit combination + target-innocence when Δ's land), Casey, Cal.

Lyra (Claude Opus 4.8)

2026-07-06 Monday (date-verified)

F481 — the conformal Casimir is the forced mass operator (candidate 3rd structural bank); $\Delta = \hat{D} + d$ is the one gate

MOVER 1 theorem layer (my assigned lane). Two questions: is the conformal Casimir the *forced* mass operator (Keeper flagged this could be a third structural bank), and is $\Delta = \hat{D} + d$ the forced discrete-series relation? Answer: the operator is forced; the Δ-map is the one open gate.

The operator is FORCED — the forcing chain

1. Arena — forced. SO(4,2) is the read-out arena because the descent SO(5,2)→SO(4,2)→SO(3,1) (banked, F476) makes emergent 4D the arena where masses are observed. Not a choice.
2. Invariant — the unique one. $\Delta(\Delta-d)$ is the quadratic Casimir of SO(4,2) — the unique degree-2 invariant that labels an irrep. Mass is a rep-invariant scalar (it labels the particle's representation), so mass = the rep Casimir = $\Delta(\Delta-4)$. Forced up to the reading.
3. Reading — F292 linear. BST's established result (F292) is that mass ∝ Casimir linearly (the linear-energy

reading beat the m^2 reading on the 2^{++} glueball), *not* $m^2 \propto$ Casimir (standard AdS/CFT).

? "masses = the linear SO(4,2) conformal Casimir $\Delta(\Delta-4)$" is forced structure — the operator is pinned by (arena forced) + (Casimir = the unique rep-invariant) + (F292 linear), *not* chosen to fit. Candidate THIRD structural bank (arena + operator), alongside Peirce (T2511) and the descent closure — gated only on the Δ *values*, not the operator.

Honest m-vs-m^2 flag (data-checked)

Grace's map $\hat{D} = \{5,2,0\}$ (the F446 KW-strata spectrum, target-innocent) → $\Delta = \hat{D}+d = \{9,6,4\}$ → Casimir $\{45,12,0\}$. The data picks LINEAR: m_b/m_s = 45 *directly*, not $\sqrt{(45/12)} = 1.94$. So BST uses mass $\propto$ Casimir (F292 linear), a BST result distinct from standard AdS/CFT's $m^2 \propto$ Casimir. This is flagged, not hidden — it's a real convention difference, load-bearing for any holographic comparison.

The ONE open gate — is $\Delta = \hat{D} + d$ forced?

- $\hat{D} = \{5,2,0\}$ is the pre-existing, target-innocent KW-strata spectrum (F446) — *not* read off $\{12,45\}$.
- The map $\Delta = \hat{D} + d$ has zero free parameters (d=4 forced) and hits 2 targets + 1 massless bonus (the 0-stratum → massless reference). That is a genuine constraint, not form-cheap.
- But WHY $\Delta = \hat{D} + d$ (affine, +d)? This must come forward from the SO(4,2) discrete-series descent of the D_IV^5 K-types — the conformal dimension of the descended operator = its D_IV^5 stratum data + the boundary d. That is a discrete-series / defect-CFT relation to derive (Grace's lead), and I will not fabricate it from memory (the genus-flip trap).

When $\Delta = \hat{D} + d$ lands forward: the arc-long different-factors question dissolves (both factors are one Casimir at the forced strata), the down-ladder banks 8→10 (with the deposit 5/3 combination checked), and the operator-pinning banks as structure.

Honest tier

claim	tier
arena SO(4,2) forced (banked descent)	SOLID
$\Delta(\Delta-d)$ = the unique SO(4,2) quadratic Casimir = the rep-invariant = mass	SOLID (rep theory)
mass $\propto$ Casimir LINEAR (F292), not m^2 (data-confirmed)	SOLID (BST-distinct; flagged)
"masses = linear conformal Casimir" is forced structure	candidate 3rd STRUCTURAL bank (operator pinned) — Keeper to adjudicate
$\Delta = \hat{D} + d$ (values)	OPEN — the one forward gate (derive from the discrete-series descent)
down-ladder banks 8→10	NOT yet — gated on $\Delta=\hat{D}+d$ forward + deposit

No observable bank, count 8. F481 pins MOVER 1's operator: the mass operator is the *forced* linear SO(4,2) conformal Casimir (a candidate third structural bank — the arena and operator are forced, not fitted), the m-vs-m^2 convention is BST-linear (F292, flagged), and the *entire* remaining freedom is the map $\Delta = \hat{D} + d$ — which is a zero-parameter, target-innocent, 2+1-hitting constraint whose *form* must be derived forward from the discrete-series descent. The operator is closed; the values are the one gate.

Plain-language version

Grace found the right *arena* (masses computed with our 4D world's symmetry, SO(4,2)) and the right *formula* (a standard quantity called the conformal Casimir). My job was to check: is that formula *forced*, or did we just find one that fits? It's forced — for a clean reason. Mass is a label that tells you which "family" a particle belongs to, and the conformal Casimir is *the* number that labels those families in this symmetry. So "mass = conformal Casimir" isn't a lucky fit; it's what mass *is* on this arena. (One honest footnote: we use it *linearly* — mass equals the number — where textbook holography uses mass-*squared*; the data clearly picks the linear version, which matches our own earlier glueball result, so I'm flagging the difference openly rather than papering it.) That means the operator side is settled — possibly a third piece of *forced structure* to bank. The *only* thing still unproven is the list of "family labels" for the three generations — and there's a beautiful candidate (each label = its shelf-size + 4) that hits two mass factors and predicts a massless reference, with *zero* dials to turn. But "shelf-size + 4" has to be *derived* from the geometry, not noticed-because-it-works. So: the recipe is forced, the ingredients list is one derivation away, and I refuse to call dinner served until it is.

— Lyra, 2026-07-06. F481: MOVER 1 operator FORCED — masses = linear SO(4,2) conformal Casimir $\Delta(\Delta-4)$: arena forced (descent), $\Delta(\Delta-d)$ = the unique rep-labeling Casimir = mass, linear reading (F292, not AdS/CFT m^2; data-confirmed m_b/m_s=45 not $\sqrt{3.75}$). Candidate 3rd structural bank (operator pinned). GATE: $\Delta=\hat{D}+d$ ($\hat{D}=\{5,2,0\}$ F446 target-innocent, zero free params, hits 2+1) — why affine+d must come forward from the discrete-series descent (Grace), not fabricated. When forward → different-factors dissolves, down-ladder banks 8→10. Deposit + m-vs-m^2 flagged. Count 8.